

CSCE 647 IMAGE SYNTHESIS · MODULE 02 · LECTURE NOTES

The Rendering Equation

One line, published in 1986, with no closed-form solution.

MODULE OUTCOMES

- State the rendering equation and explain every term.
- Explain why its self-reference is the whole difficulty.
- Write it in operator form and expand it as a Neumann series.
- Interpret the k-th term of that series as the paths of length k.
- Convert between the hemispherical and area forms and say what each is for.
- State what the equation omits.

1 The equation

James Kajiya's 1986 paper, *The Rendering Equation*, did something unusual: it did not propose an algorithm so much as identify what every rendering algorithm had been approximating without saying so. Once written down, the existing methods could be classified by which terms they kept.

$$L_o(p, \omega_o) = L_e(p, \omega_o) + \int_{\Omega} f(p, \omega_i, \omega_o) L_i(p, \omega_i) |\cos\theta_i| d\omega_i$$

Term	Meaning	Where it comes from
$L_o(p, \omega_o)$	Radiance leaving point p in direction ω_o .	The unknown. This is what we are solving for.
$L_e(p, \omega_o)$	Radiance emitted by p itself.	Given by the scene. Zero everywhere except on light sources.
$f(p, \omega_i, \omega_o)$	The BRDF at p.	The material model (Module 07).
$L_i(p, \omega_i)$	Radiance arriving at p from direction ω_i .	This is L_o at whatever surface is visible along ω_i. Hence the difficulty.
$ \cos\theta_i $	Cosine of the angle between ω_i and the normal.	Projected area (Module 01). Geometry, not material.
$\int_{\Omega} \dots d\omega_i$	Integral over the hemisphere above p.	2π steradians of incoming directions.

THE SELF-REFERENCE IS THE WHOLE PROBLEM

Look at the fourth row. L_i is not an input to the equation — it is L_o evaluated at some other surface point, which is itself given by the same equation.

The unknown appears on both sides, inside its own integral. Mathematically this is a **Fredholm integral equation of the second kind**, and for any scene more interesting than a sphere in a void it has no closed-form solution.

Physically, the self-reference is global illumination: how bright a wall is depends on how bright the floor is, which depends on how bright the wall is. Every surface's appearance depends on every other surface's, simultaneously.

Every rendering algorithm ever published is a method for approximating this one integral. Radiosity, path tracing, photon mapping, ambient occlusion, light probes, and screen-space global illumination differ only in how they approximate it and which terms they are willing to discard.

2 Operator form and the Neumann series

The equation becomes tractable when written abstractly. Define the **transport operator** T , which takes a radiance distribution over the scene, reflects it once off every surface, and propagates the result. T is linear, because light is linear: doubling every source doubles every result, and the contribution of two sources is the sum of their individual contributions.

$$L = L_e + T L$$

Formally this rearranges to $L = (I - T)^{-1} L_e$, which is useless as written — inverting T means solving the original problem. But the inverse has a series expansion:

$$L = L_e + T L_e + T^2 L_e + T^3 L_e + \dots = \sum_{k=0}^{\infty} T^k L_e$$

CONVERGENCE IS GUARANTEED BY PHYSICS

A Neumann series converges when the operator is a contraction — when applying it reduces magnitude.

T is a contraction precisely because of energy conservation (Module 01): each reflection returns at most as much energy as arrived, and in any real scene strictly less. So $\|T\| < 1$ and the series converges.

This is a satisfying connection: the physical requirement that surfaces not create energy is exactly the mathematical condition that makes the series summable.

It also explains a failure mode. A BRDF that violates energy conservation makes T non-contractive. The series diverges, which in a renderer appears as an image that gets brighter the longer you run it and Russian roulette that never terminates. If you see that, look for the non-conserving material before you look anywhere else.

2.1 Reading the series as paths

Term	The path it represents	What you see in the image
L_e	camera → light	Light sources viewed directly.
$T L_e$	camera → surface → light	Direct lighting. Shadows and their boundaries.
$T^2 L_e$	camera → surface → surface → light	One bounce of indirect. Colour bleeding from a red wall onto a white floor; light reaching into shadow.
$T^3 L_e$	three intervening surfaces	Soft ambient fill. Individually subtle, collectively the difference between a render and a photograph.
$T^k L_e$	k surfaces	Diminishing geometrically, but never exactly zero. In a bright white room the tail is substantial.

THIS IS THE STEP THAT MAKES RENDERING COMPUTABLE

The Neumann series converts an integral equation — the unknown inside its own integral — into an infinite **sum of ordinary integrals**. Each term is a perfectly conventional high-dimensional integral with no self-reference.

And an ordinary integral is something Monte Carlo can estimate. So rendering becomes: generate light paths of every length, weight each one correctly, and sum.

That is the path tracer of Module 08. Everything between here and there — intersection, acceleration, sampling theory, variance reduction, BSDFs — is the machinery needed to do it efficiently.

It also explains the terminology you will meet constantly: a 'bounce limit' of 4 means truncating the series after T^4 , which is biased, and Russian roulette (Module 06) is the trick that terminates paths *without* introducing that bias.

3 The area form

The hemispherical form integrates over directions. An equivalent form integrates over the surfaces of the scene, obtained by a change of variables from solid angle to area.

$$L_o(p, \omega_o) = L_e + \int_A f(p, p \rightarrow q, \omega_o) L_o(q, q \rightarrow p) G(p, q) V(p, q) dA(q)$$

$$G(p, q) = \cos\theta_p \cos\theta_q / |p - q|^2 \quad V(p, q) \in \{0, 1\}$$

Term	Role
$G(p, q)$	The geometry term : the Jacobian of the change of variables. The two cosines foreshorten both patches; the inverse square converts area at q into solid angle at p. It is symmetric in p and q, which matters for bidirectional methods.
$V(p, q)$	Visibility : 1 if the segment between p and q is unobstructed, 0 otherwise. Trivial to write and the source of essentially all the computational cost — evaluating it once is a ray cast.

V IS WHY QUADRATURE FAILS

V is a binary function whose discontinuities lie along the silhouettes of every object in the scene, as seen from every point. The integrand is therefore discontinuous almost everywhere, in a pattern that depends on the full scene geometry.

Classical numerical integration — Simpson's rule, Gaussian quadrature, anything with a convergence proof — assumes smoothness, usually several derivatives of it. On a discontinuous integrand those guarantees evaporate and the methods converge no faster than random sampling, often slower.

Monte Carlo does not care. Its $1/\sqrt{N}$ convergence requires only finite variance, not continuity, which is a large part of why it is the method that survived (§5).

3.1 Which form to use

	Hemispherical form	Area form
Integrates over	Directions on the hemisphere.	Surface points in the scene.
Natural sampling	Sample the BRDF : choose a direction the material is likely to scatter into, and trace a ray.	Sample the lights : choose a point on a light source directly, and trace a shadow ray to it.
Visibility	Implicit — tracing the ray resolves it.	Explicit — V must be evaluated with a shadow ray.
Fails when	Lights are small or distant . Random directions almost never hit a small light, so almost every sample returns zero and the few that hit return a large value. Enormous variance.	Surfaces are near-specular . The sampled light point is almost never inside the narrow BRDF lobe, so almost every sample contributes nothing. Enormous variance.

THE COMPLEMENTARY FAILURE IS THE KEY OBSERVATION

Direction sampling is excellent for glossy surfaces and terrible for small lights. Light sampling is excellent for small lights and terrible for glossy surfaces.

The cases where each strategy fails are precisely the cases where the other succeeds. This is not a coincidence — both strategies are trying to concentrate samples where the integrand is large, and the integrand is a product of a BRDF factor and a light factor. Each strategy matches one factor and ignores the other.

Eric Veach's **multiple importance sampling** combines several strategies with weights that are provably near-optimal, achieving variance close to the better strategy in every configuration without being told which one that is.

It is the single most important technique in this course, it is about fifteen lines of code, and it exists because of the asymmetry on this page. Module 08.

4 What the equation omits

Omission	What is assumed	Consequence
Participating media	Vacuum between surfaces, so radiance is invariant along rays.	No fog, smoke, clouds, or underwater scenes. Module 10 generalises the equation to fix this.
Subsurface scattering	Light leaves a surface at the point where it entered.	Skin, marble, milk, and wax all look wrong. Requires a BSSRDF, which takes two surface points rather than one.
Polarisation	Light has no polarisation state.	Incorrect for some dielectric reflections and most sky models; visible through a polarising filter.
Fluorescence, phosphorescence	Wavelength out equals wavelength in; emission is instantaneous.	Fluorescent paints and optical brighteners — in laundry detergent, safety clothing — are unrepresentable.
Wave optics	Geometric optics: light travels in rays and does not interfere.	No diffraction, no thin-film iridescence, no structural colour. Soap bubbles, beetle shells, and CD surfaces need a different model.

'UNBIASED' MEANS CORRECT FOR THIS EQUATION, NOT FOR REALITY

Module 05 will define an unbiased estimator as one whose expected value equals the quantity being estimated. The quantity being estimated is **the value of the rendering equation for the scene as described**.

It is not the value a camera would record. The equation omits the physics in the table above, and the scene description omits far more: measured BRDFs are fitted approximations, geometry is tessellated, textures are sampled, and the light spectra are guesses.

This distinction is routinely lost, and it matters practically. When a fully converged render does not match a reference photograph, the discrepancy is usually in the model, not the integrator — and those are debugged completely differently.

Knowing which one you are debugging saves days. The test that separates them is convergence: a renderer bug typically shows as a difference that persists at any sample count *and* changes with the integrator; a model error shows as a stable, converged image that is simply not the photograph.

5 Three families of solution

Family	Approach	Where it stands
Finite element	Radiosity. Discretise surfaces into patches, assume every patch is Lambertian, compute form factors between all pairs, and solve the resulting linear system.	Largely abandoned. It handles only diffuse transport, the form-factor matrix is $O(n^2)$ in patch count, and the meshing required is itself a hard problem. Historically important and worth knowing about.
Monte Carlo	Sample light paths at random, weight each by the probability of having chosen it, and average.	Won, comprehensively. Every production offline renderer is built this way. The rest of this course.
Approximation	Compute the first terms of the series properly and replace the tail with something cheap: ambient occlusion, irradiance probes, screen-space global illumination, baked lightmaps.	The domain of real-time rendering and of CSCE 641. Not wrong — correct for a stated budget, and the budget is the design constraint.

WHY MONTE CARLO WON

Dimension independence. Monte Carlo error falls as $1/\sqrt{N}$ regardless of dimension. A path with k bounces is an integral in roughly $2k$ dimensions, and the series sums over all k , so the effective dimension is unbounded. Every deterministic quadrature rule degrades exponentially with dimension; this one does not degrade at all.

Discontinuity tolerance. Convergence needs only finite variance, not smoothness. Given what §3 said about V , this is decisive.

Point evaluation is sufficient. Monte Carlo needs only the ability to evaluate the integrand at a point, which is exactly what a ray cast provides. No derivatives, no structure, no mesh.

Unbiasedness. The estimate improves in expectation with every additional sample and converges to the right answer, so you can stop when the noise is acceptable rather than committing to a discretisation in advance. Module 05 makes all of this precise.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Kajiya — The Rendering Equation (1986, free PDF)	The original paper. Short, readable, and worth reading in the original — the framing of the problem is the contribution.
PBRT 4th ed. — Chapter 13, Light Transport I: Surface Reflection	The rendering equation, both forms, the path-integral formulation, and the derivation of the geometry term.
Veach thesis — Chapter 8, The Path Integral Formulation	The cleanest statement of the measurement equation and the path integral. Read it before Module 11; skim it now.
TU Wien Rendering — the rendering equation lectures	The Neumann-series-as-path-lengths interpretation, built up visually. Good complement to §2.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Write the rendering equation from memory, then check it. Repeat until you can produce it exactly, including which quantities carry which arguments.
- 2 Derive the area form from the hemispherical form, showing explicitly that the Jacobian of the change of variables is the geometry term.

- 3 For a scene of two parallel unit squares facing each other at unit distance, both Lambertian with albedo 0.5, one emitting, compute the first three terms of the Neumann series by hand. Report the ratio between successive terms.
- 4 Repeat the previous exercise with albedo 0.9 and report the ratio. Explain what this implies about bounce limits in bright scenes.
- 5 In a renderer or in PBRT, render a Cornell box with maximum path depths of 1, 2, 3, 5, and 16. Difference consecutive images and report the energy in each difference. Plot it.
- 6 For the same scene, measure what fraction of total image energy comes from paths of each length. Compare a white box against a dark one.
- 7 Construct a scene where direction sampling has very high variance, and another where light sampling does. Render both with both strategies and report the variance in each of the four cases.
- 8 Pick one omission from §4 and write a page on what would have to change in the equation to include it. Participating media and fluorescence are the most instructive.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 State the rendering equation and name every term and its units.
- 2 Why does the equation have no closed-form solution, and what is the physical meaning of that fact?
- 3 Write the operator form and the Neumann series, and say what guarantees convergence.
- 4 What does the k -th term of the series represent, and what does a bounce limit of 4 correspond to?
- 5 What is the geometry term, and where does it come from?
- 6 Why is the visibility term the reason classical quadrature is unsuitable?
- 7 Give the failure mode of direction sampling and of area sampling, and say why their combination is promising.
- 8 Name four things the rendering equation does not model.
- 9 What exactly does 'unbiased' mean, and how would you tell a renderer bug from a scene-model error?
- 10 Give three reasons Monte Carlo displaced radiosity.